

Regional Quantitative Problem-Solving Circle

Solution Key 4: Graph Theory Foundations (Weeks 7–8)

Lead Architect: Mohamed Jad Srifi

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Rigorous Proofs & Structural Solutions

Problem 1 (Degree-Sum & Odd-Vertex Parity — Tournament Structure). In a group of 15 quantitative researchers, is it possible for each researcher to collaborate with exactly 7 other researchers? Generalize this to prove that any graph must contain an even number of vertices of odd degree.

Proof. Let the group of researchers be the vertex set V of a graph $G = (V, E)$, where $|V| = 15$. An edge exists between two vertices if they collaborate. If each researcher collaborates with exactly 7 others, then the degree of every vertex is $\deg(v) = 7$. By the Handshake Lemma, the sum of all vertex degrees must equal exactly twice the number of edges:

$$\sum_{v \in V} \deg(v) = 2|E|$$

Substituting the known parameters:

$$15 \times 7 = 105 = 2|E|$$

Because the edge count $|E|$ must be an integer, $2|E|$ must be strictly even. However, 105 is an odd number, yielding a strict algebraic parity contradiction. Thus, such a collaboration matrix is mathematically impossible.

Generalization: For any graph $G = (V, E)$, we partition the vertex set V into two disjoint subsets: V_{even} (vertices with an even degree) and V_{odd} (vertices with an odd degree). Applying the Handshake Lemma:

$$\sum_{v \in V_{\text{even}}} \deg(v) + \sum_{v \in V_{\text{odd}}} \deg(v) = 2|E|$$

Rearranging to isolate the odd vertices:

$$\sum_{v \in V_{\text{odd}}} \deg(v) = 2|E| - \sum_{v \in V_{\text{even}}} \deg(v)$$

The right side of the equation is the difference between two even numbers (since $2|E|$ is even, and a sum of even degrees is trivially even). Therefore, the resulting difference is strictly an even integer. Because every term in the sum $\sum_{v \in V_{\text{odd}}} \deg(v)$ is an odd number, a sum of odd integers can only equal an even integer if there is an even quantity of those terms. Thus, the cardinality $|V_{\text{odd}}|$ must be even. $\square$

Problem 2 (Bipartite Characterization & Odd Cycles). Prove that an undirected graph $G = (V, E)$ is bipartite if and only if it contains no odd-length cycles.

Proof. We must prove both directions of the biconditional statement.

Forward Direction ($\implies$): Assume $G = (V, E)$ is bipartite. By definition, its vertex set V can be partitioned into two disjoint independent sets V_1 and V_2 such that every edge connects a vertex in V_1 to a vertex in V_2 . Consider any arbitrary cycle $C = v_0, v_1, v_2, \dots, v_k = v_0$ in G . Without loss of generality, let $v_0 \in V_1$. Because edges strictly alternate between the two sets, $v_1 \in V_2, v_2 \in V_1, v_3 \in V_2,$

and so forth. For any index i , $v_i \in V_1$ if i is even, and $v_i \in V_2$ if i is odd. Since the cycle must close and terminate back at $v_0 \in V_1$, the final index k must be even. The total number of edges traversed in the cycle is k , which is strictly even. Thus, G contains no odd-length cycles.

Reverse Direction ($\Leftarrow$): Assume G contains no odd-length cycles. We must construct a valid bipartite partition. Without loss of generality, assume G is connected. Select an arbitrary starting vertex $s \in V$. We partition V by extracting Breadth-First Search (BFS) distance levels. Let $L_k = \{v \in V \mid \text{dist}(s, v) = k\}$. Define V_1 as the union of all levels with an even index k , and V_2 as the union of all levels with an odd index k . To prove this partition is valid, we must show no edge exists strictly within V_1 or strictly within V_2 . Assume for the sake of contradiction that an edge $e = \{u, v\}$ exists within the same level L_k . By the definition of shortest paths, there exists a path from s to u of length k , and a path from s to v of length k . Let w be the lowest common ancestor of u and v in the BFS tree, located at level L_j where $j \leq k$. The path traversing from w to u has length $k - j$, and the path from w to v has length $k - j$. Tracing the cycle from w down to u , across the intra-level edge $\{u, v\}$, and back up to w yields a cycle of total length:

$$(k - j) + 1 + (k - j) = 2(k - j) + 1$$

Because $k - j$ is an integer, $2(k - j) + 1$ is strictly an odd number. We have constructed an odd-length cycle, generating a direct contradiction with our initial assumption. Thus, no such intra-level edge $\{u, v\}$ can exist, proving G is bipartite. $\square$

Problem 3 (Eulerian Trail Construction & Bridge Invariants). A connected graph G has exactly two vertices of odd degree, u and v , and all other vertices have even degree. Prove that G admits an open Eulerian trail starting at u and ending at v .

Proof. Let $G = (V, E)$ be a connected graph containing exactly two vertices of odd degree, u and v . All other vertices $x \in V \setminus \{u, v\}$ possess a strictly even degree. We construct a modified topological graph G' by adding a single "virtual edge" $e^* = \{u, v\}$ to the existing edge set E . Thus, $G' = (V, E \cup \{e^*\})$. In the new graph G' , the degree of u is $\deg_{G'}(u) = \deg_G(u) + 1$. Because $\deg_G(u)$ was odd, adding one makes $\deg_{G'}(u)$ strictly even. Symmetrically, $\deg_{G'}(v) = \deg_G(v) + 1$, which is also strictly even. The degrees of all remaining vertices are unchanged and are therefore still even. Because every single vertex in G' now possesses an even degree, and G' remains fully connected, standard Eulerian transition parity dictates that G' satisfies the necessary and sufficient conditions for the existence of a closed Eulerian circuit. Let C be this Eulerian circuit in G' . By definition, C traverses every edge in G' exactly once, which means it must cross the virtual edge e^* exactly once. Since C is a continuous closed loop, we can cyclically shift our traversal starting point without breaking the circuit. Let us view the traversal of C as starting exactly at vertex v , immediately traveling across the virtual edge e^* to vertex u , and then continuing continuously along the remaining edges of the graph to eventually return to v . If we now strictly remove the virtual edge e^* from this traversal sequence, the remaining sequence of edges forms an open, non-closing trail. This trail begins exactly at u , traverses every actual edge of the original graph G exactly once (since we only discarded e^*), and terminates exactly at v . Therefore, G admits an open Eulerian trail linking its two odd-degree vertices. $\square$